

CSRMS

NADV 744 · Advanced Development Systems · Group Assignment 2026
Sol Plaatje University

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The problem

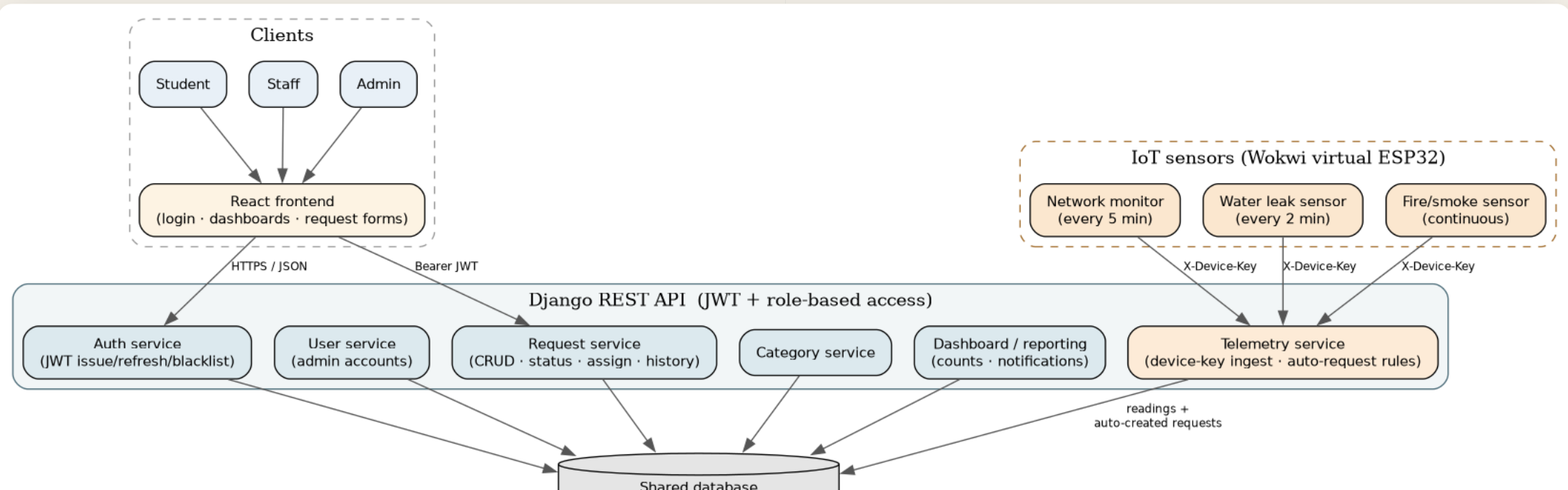
- Campus maintenance runs on WhatsApp and hallway conversations
- Reports arrive without structure: no category, no location, no owner
- Nobody can answer: *how many problems, how long to fix, which buildings?*
- Some failures **should not depend on a human noticing** — leaks and overheating happen at 2am



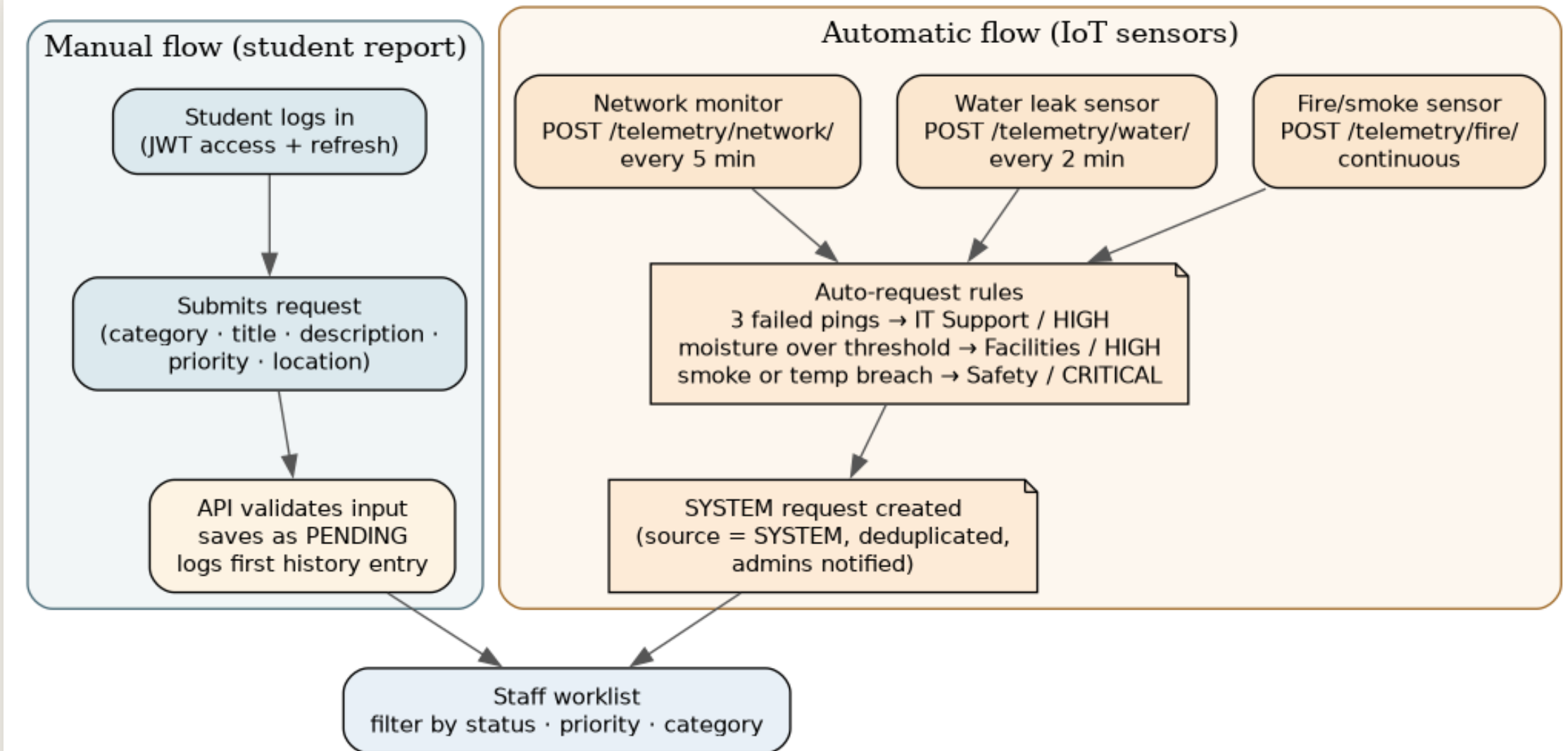
What we built

One system, two ways in:

- Students log problems → staff resolve them through a managed workflow
- Simulated IoT sensors raise requests **on their own** when conditions demand it



How a request travels



Security evidence

- Deliberate wrong-password login returns **HTTP 401** and grants no session
- A valid login returns JWT access and refresh tokens
- Students cannot read another student's request, even with a guessed ID
- A user JWT cannot post telemetry; sensors require a type-bound `X-Device-Key`
- Postman provides a visible API fallback for login, telemetry, and deduplication

Security model

Layer	Decision
Auth	JWT on every endpoint (register/login/refresh excepted)
Roles	STUDENT / STAFF / ADMIN — registration always creates STUDENT, server-side
Objects	A student who guesses another's request ID gets a 404 , proven by tests
Devices	Per-sensor keys, SHA-256 hashed, type-bound, individually revocable
Passwords	Django validators + PBKDF2

IoT auto-request rules

Sensor	Cadence	Trigger	Ticket raised
Network monitor	5 min	3 consecutive failed pings	IT Support · HIGH
Water leak	2 min	moisture > 60%	Facilities · HIGH
Fire/smoke	10 s	smoke $\geq$ 40 or temp $\geq$ 50 °C	Safety · CRITICAL

Open SYSTEM tickets are **deduplicated** per sensor+location — a stuck sensor raises one ticket, not a hundred.

Network, water, and fire/smoke triggers are verified by the backend tests and can be demonstrated live or through Postman when Wokwi or the network is unavailable.



Live demo

Student report → staff workflow → sensor-triggered ticket



Student view

CSRMS
CAMPUS SERVICE DESK

Overview

Requests

SOL PLAATJE UNIVERSITY · STUDENT WORKSPACE

Hello, Naledi.

+ Log a request

YOUR SERVICE BRIEF

Track your campus requests.

Log a problem once and follow every step the service desk takes.

OPEN REQUESTS

1

IN PROGRESS

1

RESOLVED

1

MY TOTAL

3

Latest requests

CSR-1003

Low

Broken window pane in lecture hall B

Facilities · 12 min ago

Pending

CSR-1002

High

WiFi drops out every evening in Residence A

IT Support · 12 min ago

In progress

Staff workflow with full history

REQUEST DETAIL



CSR-1002

High

In progress

WiFi drops out every evening in Residence A

IT Support · Residence A common room · logged by Naledi Khumalo · 12 min ago

Reported via the campus portal. Please inspect Residence A common room.

SERVICE DESK ACTIONS

Mark resolved

Reassign (now Lerato Mokoena)



Add a note to the timeline...

Post note

WORKFLOW HISTORY

Live sensor telemetry

CSRMS
CAMPUS SERVICE DESK

Overview

Requests

Sensors

SOL PLAATJE UNIVERSITY · STAFF WORKSPACE

Hello, Lerato.

+ Log a request



TELEMETRY SERVICE

Sensor activity

Readings posted by the three campus sensors. When a threshold is crossed the backend raises a request on its own.

Live (12 h)

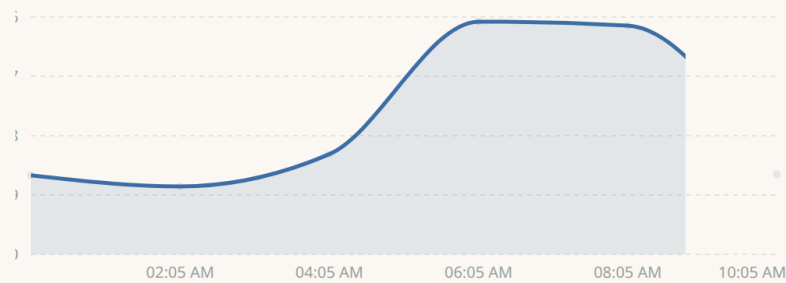
24 hours

7 days

Network monitor

Gateway latency in milliseconds

Live



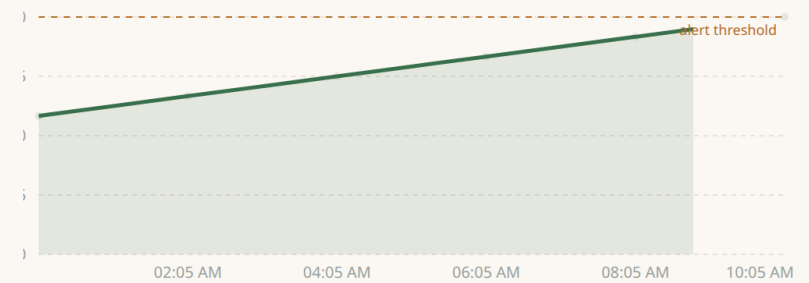
RECORDED READINGS

12.1 MS

Water leak sensor

Moisture percentage near pipes

Live



RECORDED READINGS

60 %

Fire & smoke sensor

Smoke concentration and temperature

Watched



Testing & evaluation

- **61 automated API/service tests** — every security claim has a negative test
- Firmware: structural validation + Wokwi scenarios driving sensors to their limits
- Playwright walkthrough of all three roles doubles as the screenshot generator
- Postman collection covers the API as a manual demonstration and fallback
- Performance (50 rounds/endpoint): **p95 under 8 ms** on all reads; JWT checks are DB-free, lists are paginated

Limitations & future work

Honest limits: simulated hardware · pull-only notifications · laptop-scale perf data

Next, in order:

1. Email/push notifications with staff digests
2. SLA tracking and automatic escalation
3. Real ESP32 hardware (same sketches, new WiFi credentials)
4. MySQL deployment behind HTTPS



One workflow, recorded end to end



Thank you

github.com/altechemist/NADV74

`README.md` → running everything in three commands

